

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

Anonymous authors
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ABSTRACT

Representation alignment (REPA) accelerates generative training by distilling representations from a pretrained vision encoder into intermediate diffusion features. We study which property of the target representation is most relevant for generation: its *global* semantic information (e.g., as measured by ImageNet-1K accuracy) or its *spatial* structure (i.e., pairwise cosine similarity between patch tokens). Existing practice often assumes that stronger global semantic performance yields a better alignment target for generation. We test this assumption through a large-scale empirical analysis over 27 vision encoders and multiple diffusion-model scales. The resulting evidence indicates that spatial structure, rather than global performance, is the dominant predictor of target-representation effectiveness for generation. We then introduce two minimal modifications that specifically increase the transfer of *spatial* information: replacing the standard REPA MLP projection with a lightweight convolutional projection and applying spatial normalization to the external representation. The resulting method, iREPA, requires <4 lines of code and consistently improves REPA convergence speed across diverse vision encoders, model sizes, and training variants, including REPA, REPA-E, Meanflow, and JiT. These results motivate a more mechanism-level account of representation alignment and suggest that generative training can be improved by explicitly preserving spatial structure during feature transfer.

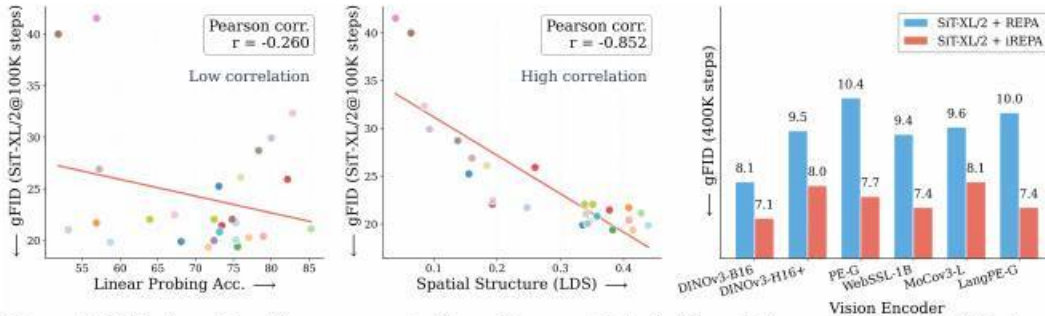


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis over 27 heterogeneous vision encoders shows that spatial structure, rather than global performance measured by linear probing accuracy, is the stronger predictor of target-representation performance for generation. *Right:* We test this mechanism by introducing two minimal modifications that increase spatial-feature transfer from the target representation to the diffusion model. The resulting approach consistently improves REPA convergence speed across diverse settings.

1 INTRODUCTION

Representation alignment has become an effective mechanism for accelerating diffusion-transformer training (Ma et al., 2024; Peebles & Xie, 2023). In this paradigm, intermediate diffusion features are constrained to match representations produced by pretrained self-supervised vision encoders, which has been shown to improve both convergence speed and final generation quality (Yu et al., 2024; Leng et al., 2025a). However, the causal mechanism by which the external representation improves



Figure 2: **Motivating examples — spatial structure matters.** Metric comparison showing a mismatch between ImageNet accuracy and generation quality. *Left:* PE-G has substantially higher validation accuracy (82.8% vs. 53.1%) but worse generation quality than SpatialPE-B (Bolya et al., 2025). *Right:* WebSSL-1B (Fan et al., 2025) likewise has stronger global performance (76.0% vs. 53.1%) but worse generation. **Spatial Self-Similarity:** Spatial structure provides a more informative predictor of generation quality than global performance. See §3 for the spatial-structure metric. All results are reported at 100K using SiT-XL/2 and REPA.

diffusion training remains insufficiently characterized. The central technical question is whether the benefit is primarily induced by transferring stronger global semantic information, typically quantified by linear probing performance, or by transferring spatial structure encoded in the relationships among patch-token representations.

Clarifying this mechanism is important because it determines how target encoders should be selected and how representation-alignment objectives should be designed. A common operating assumption is that an encoder’s utility for representation alignment should correlate with ImageNet-1K validation accuracy, which is often treated as a proxy for global semantic quality (Oquab et al., 2024; Chen et al., 2021). Under this assumption, target representations with better ImageNet performance should produce better generation (Yu et al., 2024). Consequently, improvements in the linear probing accuracy of diffusion features are often interpreted as evidence that representation alignment is effective, reinforcing a global-semantic explanation of the observed gains. The following statement from REPA (Yu et al., 2024) exemplifies this interpretation:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

This paper revisits that assumption by systematically comparing two candidate explanatory variables for representation alignment: global semantic information and spatial structure. We conduct a large-scale empirical analysis over a diverse set of vision encoders, including recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024). This analysis yields three main findings.

Higher validation accuracy does not imply better representation for generation. Contrary to the prevailing assumption, vision encoders with stronger global semantic performance, measured by ImageNet-1K linear probing accuracy, can underperform as targets for generation. For example, PE-Spatial-B is a spatially tuned model derived from PE-Core-G (Bolya et al., 2025). Although PE-Spatial-B has substantially lower patch-token validation accuracy¹ than PE-Core-G (53.1% vs. 82.8%), it produces better generation under REPA (Figure 2). Similarly, WebSSL-1B (Fan et al., 2025) has much higher global performance (76.0% vs. 53.1%) but worse generation performance. In Section §2, we show that this pattern recurs across target representations, indicating that global accuracy alone is not a sufficient criterion for selecting an alignment target.

Spatial structure not global information determines generation performance. To quantify this effect, we define several metrics for the spatial self-similarity structure (Shechtman & Irani, 2007) among patch tokens (§3). We then compute correlations between REPA generation FID, linear probing accuracy, and spatial-structure metrics across 27 vision encoders and 3 diffusion-model sizes (SiT-B, SiT-L, SiT-XL). All spatial-structure metrics correlate strongly with generation FID (Pearson $|r| > 0.852$), whereas ImageNet-1K validation accuracy has substantially weaker predictive power

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

($|r| = 0.26$). Additional experiments support the same conclusion: representations with limited global information can still provide significant REPA gains. For instance, SAM2 yields better REPA generation than encoders with $\sim 60\%$ higher ImageNet accuracy (Fig. 3). Likewise, although they are not the best targets, classical spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005) also produce nontrivial gains under representation alignment (§3).

Accentuating spatial features improves convergence performance. We operationalize this observation by introducing two minimal modifications (<4 lines of code) that increase the transfer of “spatial” information from the target representation to the diffusion model. First, we apply a spatial regularization layer that increases spatial contrast in the target representations. Second, we show that the standard MLP projection used to map diffusion features to the target-representation dimension can discard spatial information (Figure 6a), and we replace it with a lightweight convolutional projection. The resulting method, iREPA, consistently improves REPA convergence speed across variations in encoders, model sizes, and training recipes *e.g.*, REPA-E (Leng et al., 2025a) and Meanflow (Geng et al., 2025) w/ REPA.

We summarize the paper’s contributions as follows:

- We provide a large-scale empirical analysis showing that spatial structure, rather than global semantic information, is the primary factor associated with the effectiveness of representation alignment.
- We introduce a simple and efficient Spatial Structure Metric (SSM) that correlates substantially more strongly with downstream FID performance than linear probing scores.
- We propose minimal modifications that better preserve and transfer spatial information from target representations to diffusion features. The resulting method improves convergence speed over REPA across target representations, model architectures, and training-recipe variants, including REPA, REPA-E, Meanflow w/ REPA, and JiT w/ REPA.

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We first present a set of motivating empirical cases in which stronger global performance of the target representation, as measured by ImageNet-1K accuracy, does not translate into improved generation performance under REPA. We subsequently show in §3 that these observations are more coherently accounted for by metrics that characterize the spatial structure of the target representation.

Recent vision encoders. We evaluate several recent vision encoders, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a compact spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As reported in Figure 2, PE-Core-G obtains substantially higher ImageNet-1K accuracy than PE-Spatial-B (82.8% vs 53.1%), yet yields worse performance when used as the REPA target representation (FID 32.3 vs 21.0). WebSSL-1B (1.2B) exhibits the same discrepancy: although its ImageNet-1K accuracy is much higher (76.0% vs 53.1%), its REPA generation performance is lower (FID 26.1 vs 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To further isolate the limited role of global information in this setting, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024), a small model with weak global classification signal and a validation accuracy of only 24.1% (Fig. 3a). Despite this low global-performance score, SAM2-S achieves a better FID under REPA than several encoders with substantially higher ImageNet-1K accuracy *e.g.*, PE-Core-G (82.8%).

Larger models within same encoder family can have similar or worse generation performance. A common assumption is that, within a fixed encoder family, larger variants provide stronger target representations because they generally obtain higher ImageNet-1K accuracy. For representation alignment, however, larger variants frequently produce comparable (DINOv2) or even *worse* (PE, Cradio) generation performance (Fig. 3b). Notably, (Yu et al., 2024) report an analogous observation for DINOv2 and attribute it to the possibility that all DINOv2 variants are distilled from DINOv2-g and therefore share similar representations. In §3, we show that the same trend is more directly explained by differences in spatial structure.

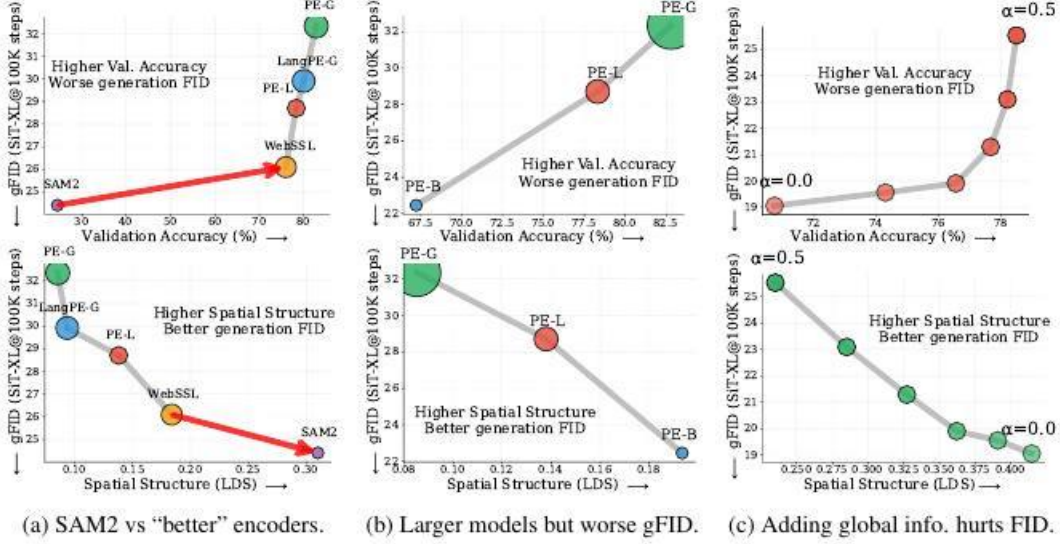


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Multiple trends show that global performance is weakly associated with generation FID under REPA. (a) SAM2-S, with validation accuracy of only 24.7%, yields better REPA generation performance than models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can have higher validation accuracy but worse generation performance. (c) Adding global information to patch tokens through CLS-token mixing improves global performance but degrades generation. **Bottom row:** Spatial structure provides a more informative generation-quality indicator than global performance. See §3 for the large-scale analysis across spatial-structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference. (a) SAM2-S, with validation accuracy of only 24.7%, yields better REPA generation performance than models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can have higher validation accuracy but worse generation performance. (c) Adding global information to patch tokens through CLS-token mixing improves global performance but degrades generation.

Adding more global information can hurt generation. To test whether injecting additional global information improves representation alignment, we conduct controlled experiments that mix the CLS token into local patch tokens, using DINOv2 as the encoder. The CLS token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. As shown in Figure 3c, increasing α from 0 to 0.5 monotonically raises linear probing accuracy from 70.7% to 78.5%. In contrast, generation quality degrades substantially, with FID increasing from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$.

These observations indicate that global representation performance is not a reliable predictor of REPA performance. We next show in §3 that spatial structure provides a more informative explanatory variable.

- **Finding 1.** Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.

3 SPATIAL STRUCTURE MATTERS MORE

We posit that the generation performance induced by a target representation is governed primarily by its spatial structure rather than by its global semantic content. This hypothesis is motivated in part by the encoder comparisons in Figure 2. To operationalize this hypothesis, we define simple metrics for the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then show that these spatial metrics correlate substantially more strongly with gFID than ImageNet-1K accuracy and that they explain the empirical discrepancies identified in §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define the following quantities:

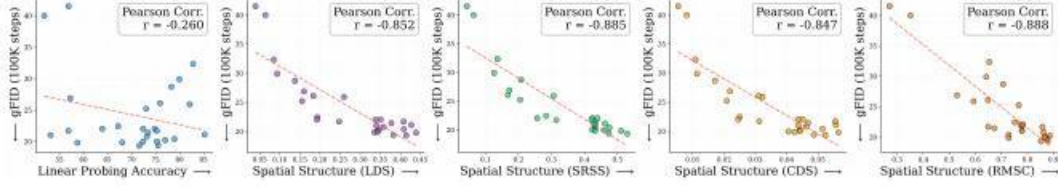


Figure 4: Spatial structure shows higher correlation with generation quality than linear probing. Correlation analysis over 27 vision encoders using SiT-XL/2 and REPA. Linear probing has weak correlation with FID (Pearson $|r| = 0.260$), whereas all spatial-structure metrics—LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$)—show substantially stronger correlation with generation performance. See Fig. 10 for encoder-labeled plots.

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ denote the extracted patch representations, with $T = H \times W$ patches. Let the kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ quantify appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. We instantiate this kernel as cosine similarity, $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ specify the spatial coordinate of each of the T tokens, and let $d(\cdot, \cdot) \in \mathbb{N}$ denote the Manhattan distance between token pairs.

We measure *spatial* self-similarity by evaluating how the token-similarity kernel $K_{\mathcal{X}}$ varies as a function of lattice distance d between patch tokens. Larger metric values correspond to stronger spatial organization, meaning that nearby patches are more similar to one another than distant patches. By default, we use a correlogram-style local-versus-distant contrast (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is computed by taking the expectation over patch representations $\mathcal{X}$. We set $r_{\text{near}} = r_{\text{far}} = H/2$ in the default configuration, and empirically find that the resulting correlations are robust to the precise threshold values. Appendix B provides exact implementation details and reports alternative spatial metrics (CDS, SRSS, RMSC), which also perform effectively (see Fig. 4).

Spatial structure correlates much higher with generation performance than linear probing. We perform a large-scale correlation analysis over 27 heterogeneous vision encoders. As shown in Figure 4, the commonly used linear probing measure has only weak association with generation quality (Pearson $|r| = 0.26$), whereas all SSM metrics exhibit much stronger correlation with generation performance (Pearson $|r| > 0.85$).

Generalization across model scales. We further evaluate whether this relationship is stable across diffusion-model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. Linear probing remains weakly correlated across scales ($|r| < 0.306$), while spatial structure maintains a substantially higher correlation with generation performance ($|r| > 0.826$).

- **Finding 2.** *Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

Spatial structure can explain previously unexplained trends. As discussed in §2, ImageNet-1K accuracy is not a predictive measure of target-representation utility for representation alignment; the spatial metrics defined above provide a more consistent explanatory signal.

(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows that PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%), but performs worse when used as the target representation for REPA (FID 32.3 vs 22.0). The spatial-structure metric resolves this mismatch: despite its lower global performance, PE-Spatial-B has substantially stronger spatial structure than PE-Core-G, which is consistent with its better generation performance.

(2) *SAM2 outperforms “better” vision encoders:* §2 shows that SAM2 has much lower ImageNet-1K accuracy (24.1%) yet yields better generation than encoders with $\sim 60\%$ higher accuracy. As shown in Fig. 3a, this performance advantage is directly aligned with SAM2’s stronger spatial structure.

(3) *Larger models in same encoder family underperform:* As shown in Fig. 3b, larger variants within the same encoder family can have higher ImageNet-1K accuracy but weaker spatial structure, which

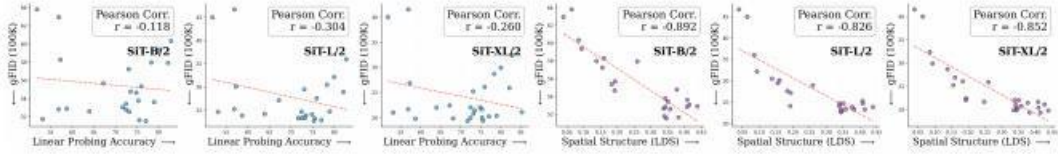
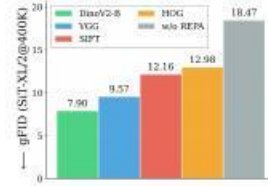


Figure 5: **Correlation analysis across model scales.** Across diffusion-model scales, spatial structure (right) consistently exhibits stronger correlation with gFID than linear probing (left).

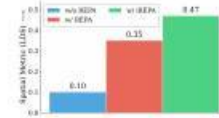
in turn corresponds to worse REPA generation performance. This explanation is also consistent with observations in Yu et al. (2024), where larger models can produce similar or worse generation results.

4) *Adding global information to patch tokens via CLS token hurts generation:* In Figure 3c, increasing the amount of global information by mixing the CLS token into patch tokens improves global performance. However, this operation reduces spatial contrast among patch tokens by increasing similarity between otherwise unrelated regions (*e.g.*, from foreground object to background). The resulting degradation in spatial structure is therefore consistent with the observed deterioration in generation performance.

If spatial structure matters more, can we use SIFT or HOG features for REPA? Surprisingly, yes. Although they are not the strongest targets, classical spatial descriptors such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014) all yield performance gains under REPA. This result provides additional evidence that representation alignment can benefit from spatially structured features even in the absence of strong global semantic information.



Can we use spatial metrics to explain gains with REPA? Yes. The proposed spatial metrics account not only for the gains obtained by REPA, but also for the improvements produced by the enhanced training recipe, iREPA, which we introduce next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

To further test the role of spatial structure in representation alignment, we modify the standard REPA training recipe with two minimal architectural and normalization changes. Both changes are designed to increase the fidelity with which spatial information is transferred from the teacher vision encoder to the student diffusion transformer.

Convolutional projection layer instead of MLP. Standard REPA uses a 3-layer MLP projection to map diffusion features into the dimensionality of the external representation. As shown in Figure 6a, this projection can be lossy with respect to spatial contrast between patch tokens. We therefore replace the MLP with a lightweight convolutional layer (kernel size 3, padding 1) applied directly on the spatial grid. The convolutional parameterization preserves local neighborhood relations through its spatial inductive bias.

Spatial normalization layer. Consistent with Siméoni et al. (2025), we observe that patch tokens from pretrained vision encoders contain a substantial global component. This is indicated by the high linear-probing scores obtained from the mean of patch tokens (Figure 14). Figure 6b further shows that mixing this global component, represented by the mean patch token, into individual patch tokens improves global performance but decreases pairwise spatial contrast. Consequently, tokens from semantically distinct regions, such as foreground objects and background, can exhibit high similarity.

Given the results in §3, we hypothesize that partially suppressing this global component can increase spatial contrast among patch tokens. The resulting pairwise-similarity structure should provide a stronger spatial signal for REPA and therefore improve generation performance. We implement this operation by applying a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

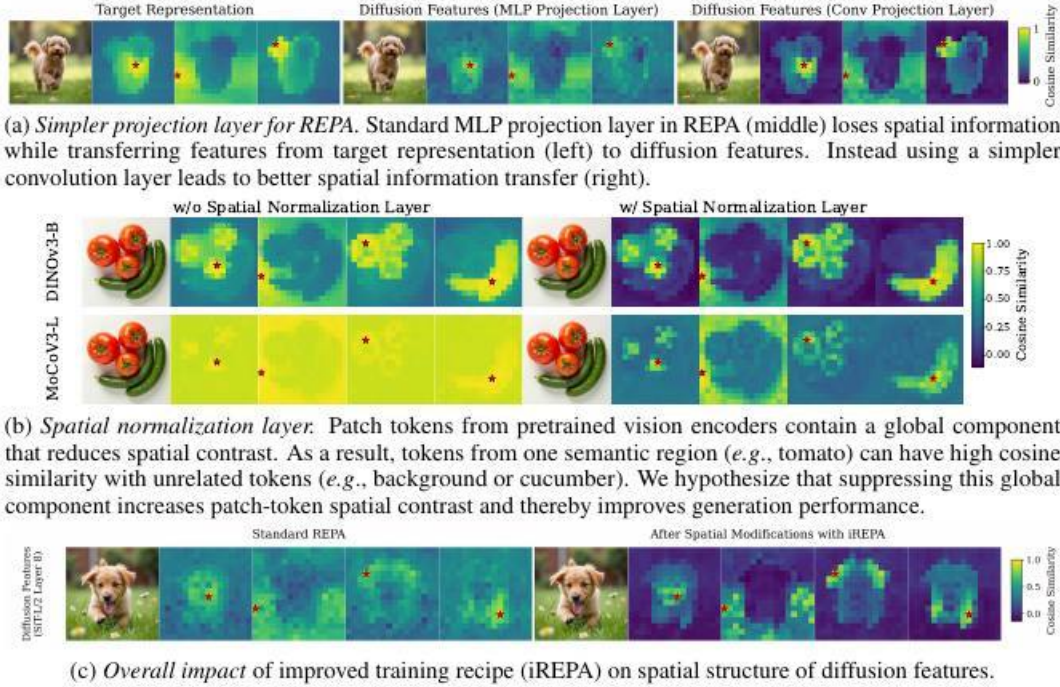


Figure 6: Two minimal iREPA modifications designed to increase spatial-feature transfer from the target representation to diffusion features. All results are reported with SiTL/2 using REPA at 100K.

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ denotes the patch-token tensor, the expectation and variance are computed along the spatial dimension, and $\epsilon = 10^{-6}$ provides numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

4.1 EXPERIMENTS

We evaluate the improved training recipe through extensive experiments on Imagenet 256×256 . The evaluation is organized around the following technical questions:

- Does iREPA consistently accelerate diffusion-transformer convergence relative to REPA across heterogeneous external representations? (Figure 7, 8, 12, 13, 15)
- Does iREPA scale with model size and remain effective under variations in the training configuration? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA transfer to recent representation-alignment methods such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025), as well as to pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We measure the convergence behavior of iREPA across multiple target encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and diffusion-model sizes (SiT-XL/2 and SiT-B/2). The results in Fig. 7 show that iREPA consistently reaches lower FID faster than baseline REPA across both target-representation and model-scale variations.

Target representation. We evaluate the generalization of iREPA across vision encoders in Fig. 8 and Table 4. iREPA improves generation quality for all evaluated encoders. Additional comparisons over the full set of 27 encoders are reported in Appendix C, E.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Re. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60
+iREPA	76.3	17.6	6.36	0.65	0.61
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59
+iREPA	70.8	19.5	6.35	0.64	0.61

Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Re. $\uparrow$
SiT-B	27.5	49.50	7.00	0.46	0.59
+iREPA	34.1	43.37	7.87	0.50	0.60
SiT-L	55.7	24.10	6.25	0.62	0.60
+iREPA	66.9	20.28	6.14	0.63	0.62
SiT-XL	70.3	19.06	5.83	0.65	0.61
+iREPA	77.9	16.96	6.26	0.66	0.61

Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Re. $\uparrow$
Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	41.7	36.0	6.61	0.52	0.62
Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	42.4	35.3	6.90	0.53	0.62
Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	38.3	38.9	7.19	0.51	0.62

(a) Vision Encoder size (b) Model size (c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA provides consistent gains over baseline REPA across diverse training settings. Unless otherwise specified, results use Dinov2 as the encoder, SiTXL/2, 100K training steps, and vanilla REPA as the baseline.

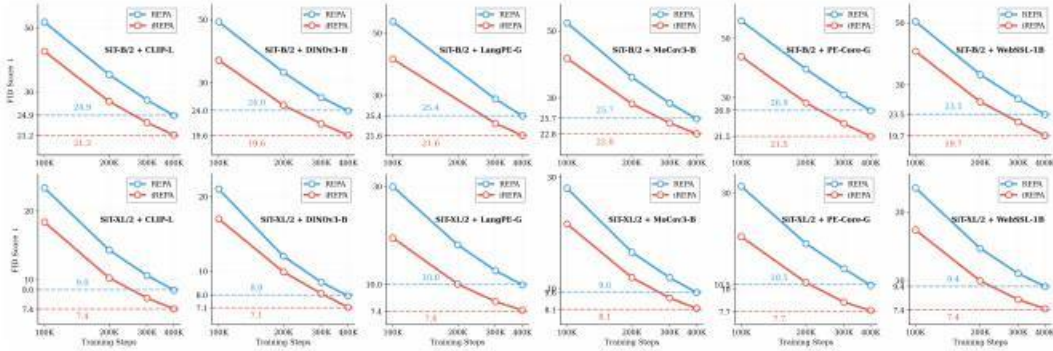


Figure 7: **Accentuating spatial features helps consistently improve convergence speed.**

Encoder size. We test whether iREPA remains effective as the target-encoder capacity changes. Table 1a reports results for PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves performance for every encoder size, and the relative improvement increases with encoder capacity (22.2% for PE-B, 38.8% for PE-L, 39.6% for PE-G).

Scalability. We analyze scalability with respect to diffusion-model size in Table 1b. The spatially oriented modifications consistently improve performance, and the relative gains are larger for larger models, indicating that the benefit of improved spatial transfer scales with model capacity.

Encoder depth. Table 1c evaluates iREPA under different alignment depths. All entries use SiT-B/2 trained for 100K iterations with DINOv3-B as the target representation. Across alignment depths, iREPA consistently improves over baseline REPA.

Abalation on different components. We isolate the contribution of each iREPA component in Table 2. Both spatial normalization and the convolutional projection layer independently improve generation quality relative to baseline REPA, and their combination produces the best result.

Training recipe. We next evaluate whether iREPA transfers across training recipes, including REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). Table 3 shows that the spatial modifications yield convergence gains across these recipes.

Classifier-free guidance. We evaluate iREPA generation quality with classifier-free guidance (CFG) in Table 4, which reports results across vision encoders after 400K training iterations. Across encoders, iREPA accelerates convergence both with and without classifier-free guidance.

Pixel-space diffusion (JiT). We also evaluate iREPA on pixel-space diffusion models, using JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA within JiT (Li & He, 2025b). The results show that emphasizing spatial-information transfer consistently accelerates REPA convergence across multiple vision encoders (e.g., DiNOv2, DiNOv3, PE etc.).

- **Finding 3.** Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).

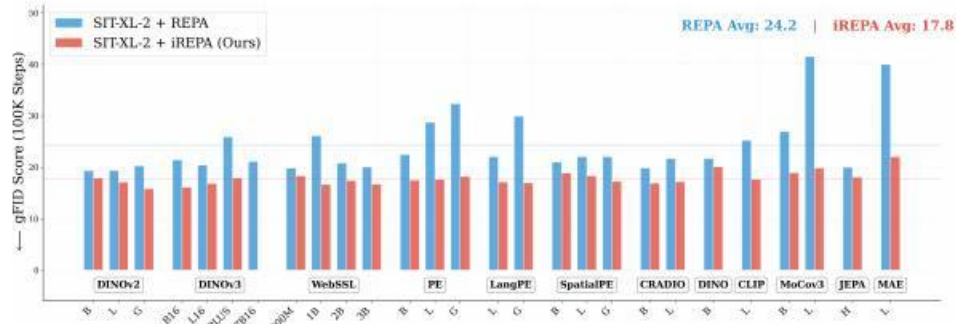


Figure 8: **Variation in target representation.** Across all 27 vision encoders, increasing spatial-feature transfer from the target representation to diffusion features with iREPA produces consistent convergence-speed improvements. See Appendix C for additional results across diverse settings.

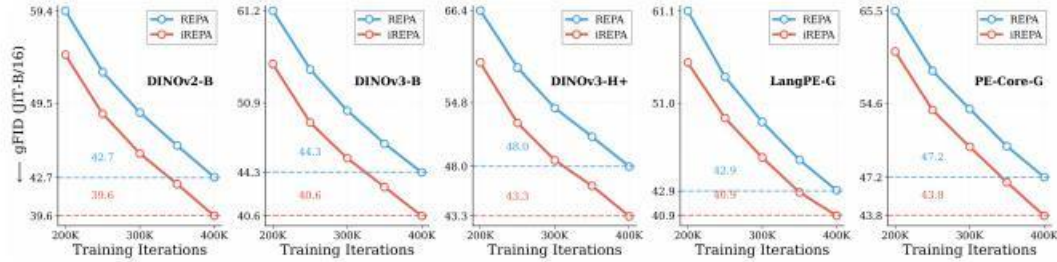


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves compare REPA and iREPA within pixel-space JiT (Li & He, 2025b) across multiple vision encoders. Increasing spatial-structure transfer consistently improves convergence speed for pixel-space diffusion models. All results are reported with JiTB/16 (Li & He, 2025b), batch size 256, and no classifier-free guidance. See Table 10 for additional results.

5 RELATED WORK

We summarize the most relevant prior work in this section and provide a more detailed discussion in Appendix H.

Representation alignment for generation. A growing body of work investigates the use of external representations to improve diffusion-model training (Pernias et al., 2023; Fuest et al., 2024). In particular, recent methods (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) demonstrate that substantial performance gains can be obtained by aligning internal diffusion features with clean-image features extracted from a pretrained vision encoder. Subsequent work extends this alignment principle to video generation and 3D generation (Zhang et al., 2025; Wu et al., 2025), and (Ma et al., 2025a) shows that representation alignment can also improve unified-model training. Despite these empirical gains, the mechanism by which self-supervised visual features facilitate diffusion training remains under-specified. Our work isolates which property of the target representation is most relevant for generation and uses that analysis to derive an improved training recipe.

Spatial vs global information tradeoff in pretrained vision encoders. Recent studies examine the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued self-supervised training can increase similarity between the global CLS token and patch tokens, which can degrade performance on dense spatial tasks. Other work explicitly trains spatially tuned models for dense prediction settings (Bolya et al., 2025; Heinrich et al., 2024). In contrast, our analysis focuses on generative training and shows that, for representation alignment, the spatial structure of a vision encoder is more predictive than its global information. This finding suggests that future target-encoder selection and training should account explicitly for spatial structure when optimizing external representations for generation.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** Across encoders, both the spatial normalization layer and the convolutional projection layer (§4) yield significant convergence-speed gains, with the best results obtained by combining them. All results are reported at 100K steps with SiT-XL/2.

Encoder	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** We apply the spatial modifications on top of REPA-E (a) and Meanflow with REPA (b), obtaining consistent gains. For Meanflow, results are reported at 1 and 4 NFEs, with CFG value 2.0. All results are reported at 100K training iterations.

6 CONCLUSION

This paper examines which property of a target representation determines the effectiveness of representation alignment: global information or spatial structure. Through a large-scale empirical analysis, we find that spatial structure, rather than global information, is the primary factor associated with REPA performance. We further validate this mechanism by introducing two simple modifications that increase the transfer of spatial information from the target representation to diffusion features. The resulting method, iREPA, consistently improves REPA convergence speed across variations in vision encoders, model sizes, and training recipes. These results motivate future work on the mechanisms of representation alignment and on target representations that are explicitly optimized for spatially faithful generative training.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+IREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+IREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+IREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+IREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+IREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+IREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+IREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+IREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across encoders with and without classifier-free guidance.** SiT-XL/2 is used as the base model. See Appendix E for detailed SiT-B/2 and SiT-L/2 results across all encoders.

REPRODUCIBILITY STATEMENT

We report the implementation details and hyperparameter settings required to reproduce the experiments in Appendix G. Code, model checkpoints, and analysis artifacts will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

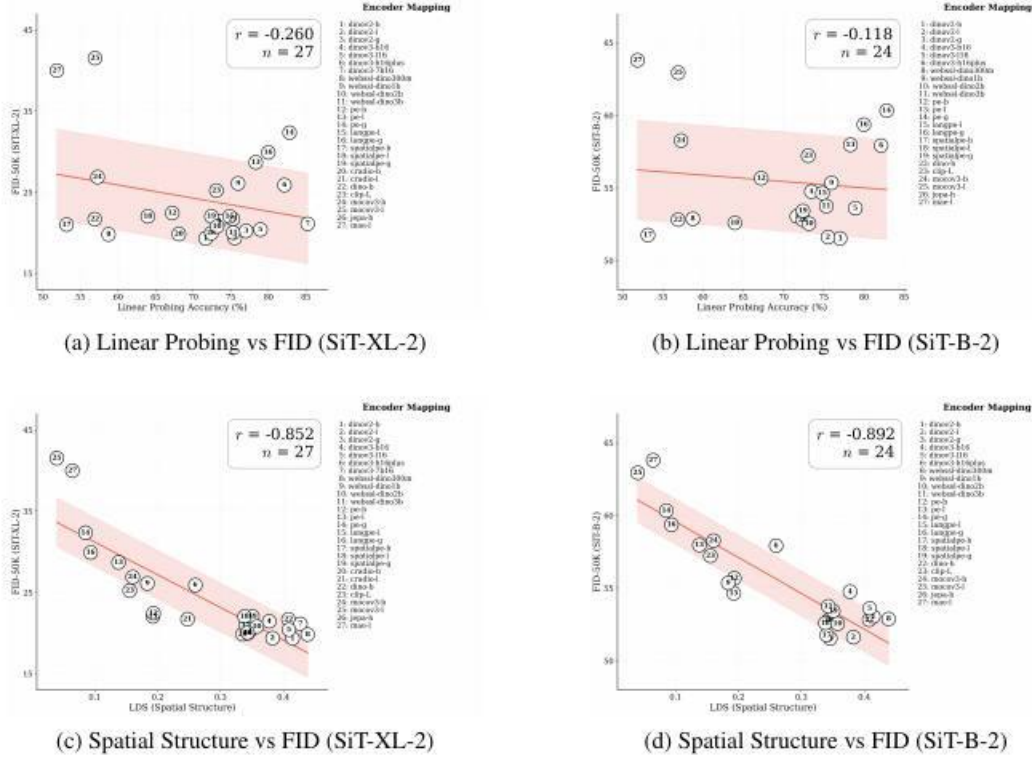


Figure 10: Detailed correlation analysis with encoder labels. The top row reports the correlation between linear probing accuracy and FID for SiT-XL-2 and SiT-B-2. The bottom row reports the corresponding correlation between spatial structure (LDS) and FID. Each point is annotated with its encoder index (see legend). Across both model scales, the spatial-structure metric exhibits substantially stronger association with generation quality.

B SPATIAL SELF-SIMILARITY METRICS



Figure 11: **Semantic region self-similarity (SRSS) metric.** Visual description of SRSS: for each image, we compute a SAM2 mask and sample anchor-positive-negative triplets. The metric measures the cosine-similarity difference between anchor-positive pairs within the mask and anchor-negative pairs outside the mask. Larger values indicate that patches from the same semantic region are more similar than patches from unrelated regions, corresponding to stronger spatial-structure preservation.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

We next define several simple metrics that quantify the spatial self-similarity structure of the target representations.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are hyperparameters. By default, we set $r_{\text{near}} = r_{\text{far}} = H/2$, and we find that the correlations are robust to the exact threshold choices. Larger values indicate that, *on average*, nearby patches are more similar than distant patches.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values indicate faster similarity decay with distance, suggesting stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we first sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object using SAM2 (Ravi et al., 2024) and downsampled to $H \times W$. We then select three types of points: anchor, positive, and negative. The anchor and positive points are sampled from within the mask (should have similar semantics), while the negative point is sampled from outside the mask (less related to anchor). Conceptually, if the encoder feature representation preserves the spatial structure, the anchor and positive points should have higher cosine similarity, while the anchor and negative points should have lower cosine similarity. Thus, we define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

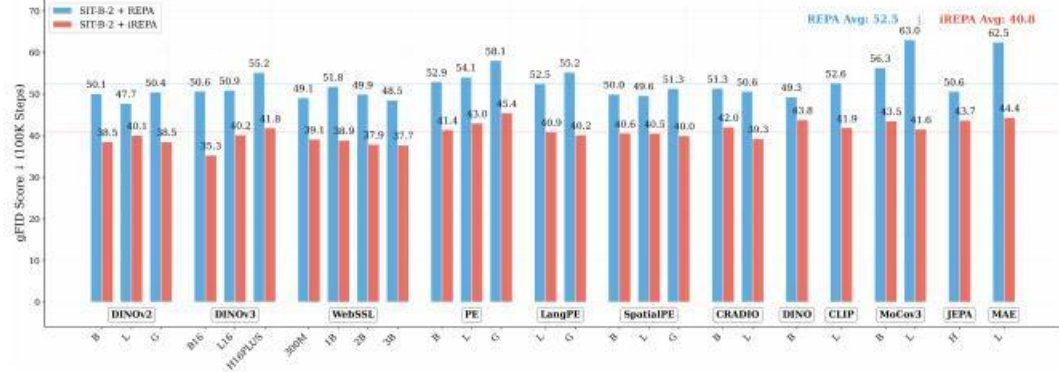
where positive points lie within Manhattan distance $d \leq r_{near}$ from the anchor inside the mask, and negative points lie at distance $d \geq r_{far}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we use a contrast-based metric to quantify spatial diversity among patch-token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

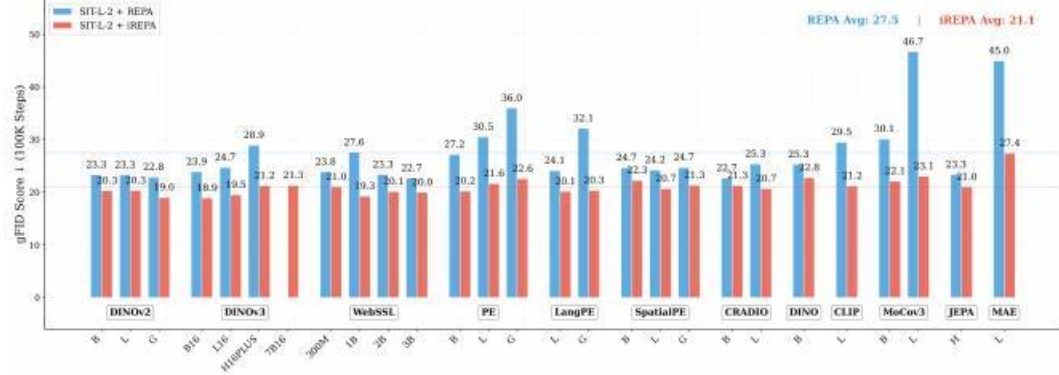
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of normalized features across all spatial locations. Higher RMSC values indicate greater spatial diversity in the feature representations and therefore stronger preservation of spatial structure. Lower values indicate more uniform feature distributions, which is consistent with reduced spatial structure when global information diminishes patch-token contrast (Siméoni et al., 2025).

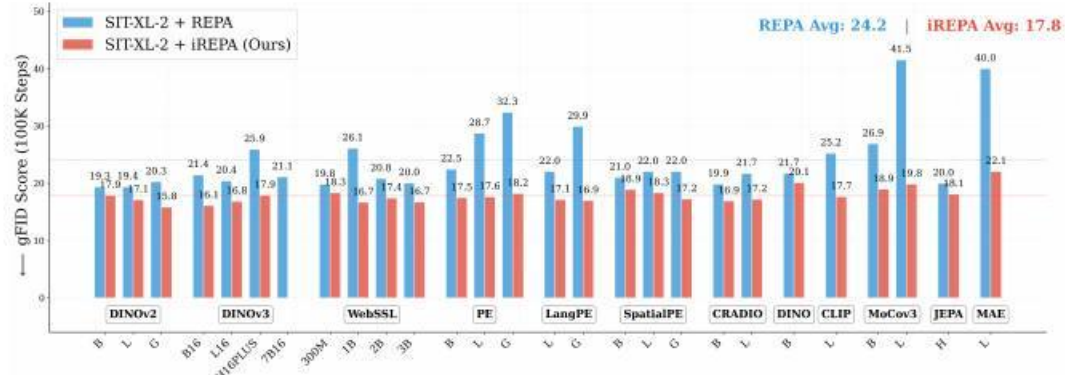
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



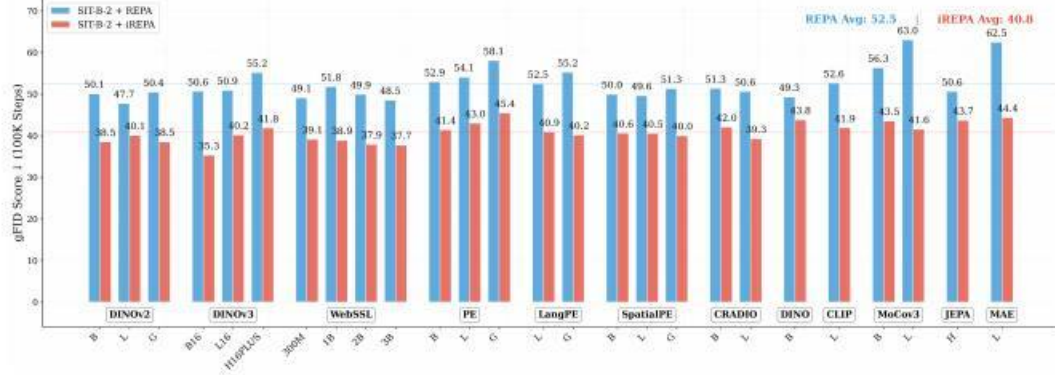
(b) SiT-L-2 (458M parameters)



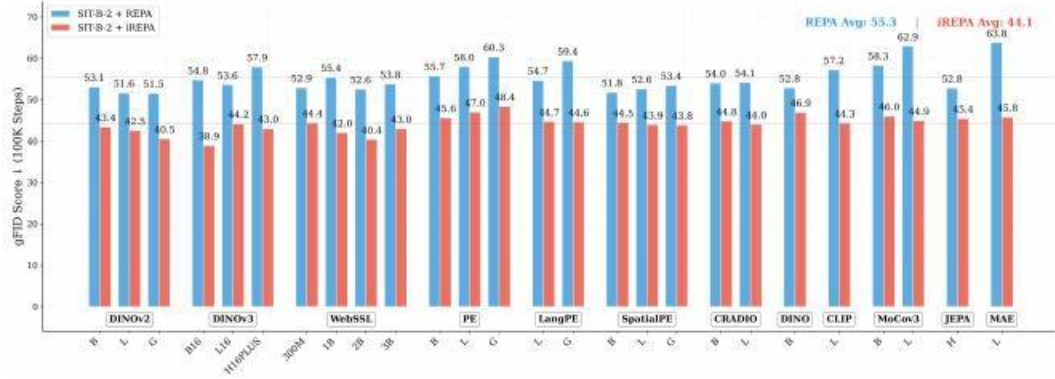
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies showing iREPA improvements across different vision encoders for SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently improves generation quality across all encoders and model sizes.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** Comparison of SiT-B-2 performance with different encoder depths (6 vs 8 layers) across various vision encoders. The results show consistent improvements with iREPA regardless of encoder depth, with slightly better performance observed with 8 encoder layers. Both configurations demonstrate significant FID improvements when spatial regularization is applied.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality across all encoders, with particularly strong gains in FID and IS metrics. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

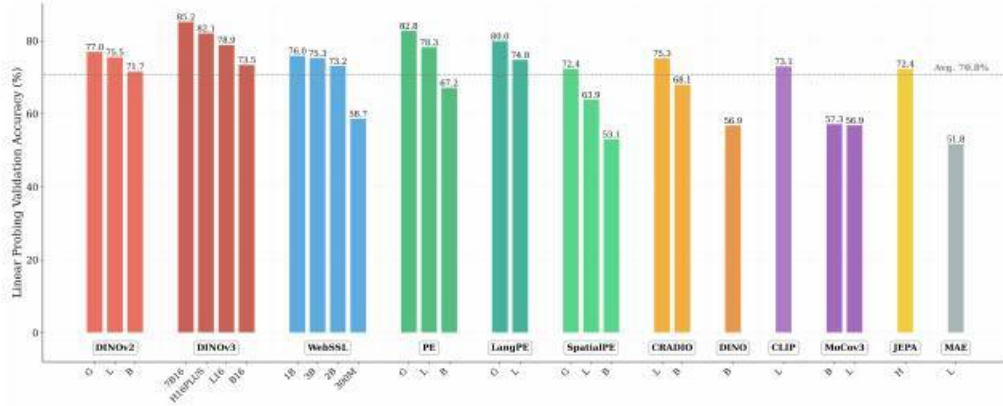


Figure 14: **Global information in mean patch tokens.** In addition to the [CLS] token, the mean of patch tokens contains substantial global semantic information. Although this component improves global performance, it reduces contrast between individual patch tokens and can therefore hinder spatial-structure transfer. Removing part of this global component through the patch-token mean improves spatial transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Comparison of REPA and iREPA under CFG (scale 2.0) across vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA consistently improves both IS and FID across all encoders.

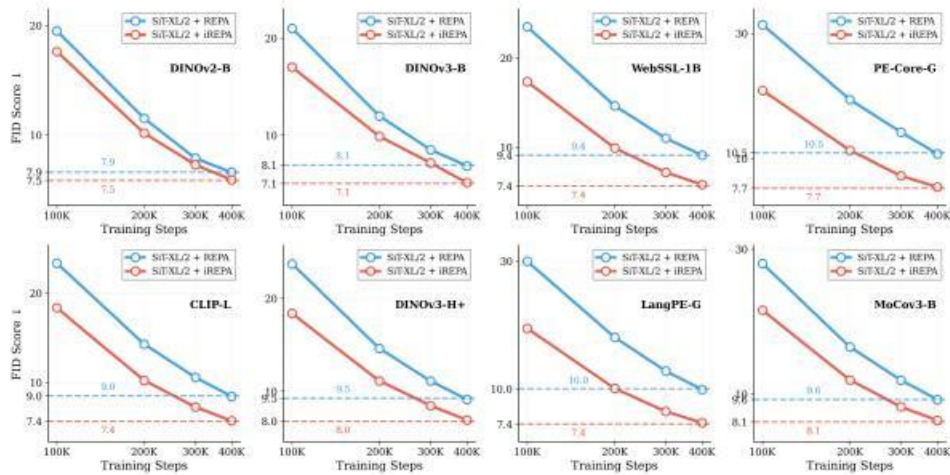


Figure 15: **Accentuating spatial features helps consistently improve convergence speed.** Results for SiT-XL/2 with REPA and iREPA.

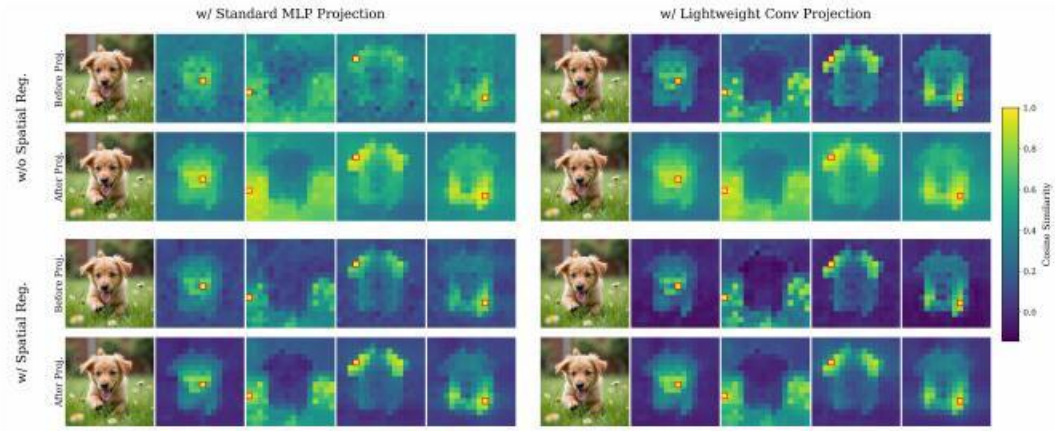


Figure 16: **Qualitative effect of two spatial-transfer modifications.** **First**, the standard MLP projection layer (top-left) loses spatial information when mapping features between the pretrained encoder representation and diffusion-model representation. A simpler convolutional projection better preserves spatial information during this transfer (top-right). **Second**, vision-encoder features often have limited spatial contrast (§2). Consequently, tokens from one semantic region (*e.g.*, dog) can exhibit high cosine similarity with unrelated tokens (*e.g.*, background). The spatial regularization layer increases spatial contrast in the learned representation, which improves generation performance. The strongest results are obtained when both modifications are used together (Table 2).

G IMPLEMENTATION DETAILS

Training setup. We follow the same setup as in REPA (Yu et al., 2024) unless otherwise specified. All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the protocol from ADM (Dhariwal & Nichol, 2021), where each image is center-cropped and resized to 256×256 resolution. We use `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout our diffusion training and inference. For spatial normalization layer we use $\gamma \in [0.6, 0.8]$ and for projection layer we use a convolutional layer with kernel size 3 and padding 1. For optimization, we use AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} , and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` to accelerate training, and gradient clipping and exponential moving average (EMA) to the generative model for stable optimization.

For REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementation. For MeanFlow (Geng et al., 2025) experiments, we received the implementation (including REPA) from original authors and adapt it to introduce the two straightforward changes for iREPA (§4).

Evaluation. For image generation evaluation, we strictly follow the ADM setup (Dhariwal & Nichol, 2021). We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), measured on 50K generated images. For sampling, we follow the approach in REPA (Yu et al., 2024), using the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we use 50 inference steps following official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been a central component of computer vision. Classical methods such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) provide robust local descriptors for tasks ranging from object detection to image matching. Although these handcrafted features are designed to capture local spatial patterns and geometric invariances, recent generative-modeling work has primarily emphasized representations from modern self-supervised encoders with strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our results support a different interpretation: because spatial-structure preservation is critical for generation quality, classical spatial features can also improve diffusion training when they are appropriately aligned. This suggests that effective target representations may be drawn from the broader class of spatial feature extractors, from handcrafted descriptors to modern learned encoders, provided that they preserve spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models in several capacities: as discriminators to accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and more recently as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have not only improved convergence speed but also enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation leverages pretrained encoders to guide student networks toward higher-fidelity samples, showing clear improvements in perceptual quality. In particular, REPA (Yu et al., 2024) aligns diffusion features with external encoders, demonstrating that representation alignment can improve both generation convergence and quality. Building on this direction, we focus not merely on alignment but on preservation of spatial structure rather than discriminative capability alone.

Denoising transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, with various formulations including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent architectures like GenTron (Chen et al., 2023) scale transformers to over 3B parameters for text-to-image synthesis. U-ViT (Bao et al., 2023) demonstrates that pure transformer backbones without U-Net inductive biases can achieve competitive performance. ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that combine denoising with sequential generation, enabling flexible

trade-offs between quality and speed. Despite these architectural advances, training such models from scratch remains computationally expensive and often requires millions of iterations to reach strong generation quality. Representation-alignment methods such as REPA (Yu et al., 2024) demonstrate that pretrained features can substantially accelerate convergence, but the mechanism responsible for this improvement has remained unclear. Our analysis indicates that the key benefit is associated with preserving spatial structure rather than only improving semantic alignment, which helps explain why certain encoders provide stronger acceleration and informs the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denosing as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. These works establish denoising as a fundamental self-supervised task that naturally encourages learning of robust features. Our findings complement this view by showing that when diffusion models are aligned with strong spatial representations from self-supervised encoders, both the generative and discriminative capabilities improve, suggesting that spatial structure preservation is a key factor in this synergy.

Scaling self-supervised vision encoders. Recent work has substantially scaled self-supervised vision models in both parameter count and dataset size. DINOv3 (Siméoni et al., 2025) trains a 7B-parameter ViT on 1.7 billion unlabeled images by aligning representations from different augmentations. WebSSL (Fan et al., 2025) shows that vision models trained on more than 2 billion images can match language-supervised models such as CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) combines multiple teacher models through distillation to produce versatile encoders across visual domains. I-JEPA (Assran et al., 2023) studies predictive architectures that learn by predicting masked regions in representation space rather than reconstructing pixels. SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations on dense prediction tasks. Although these models are commonly evaluated on global tasks such as image classification, our results show that discriminative benchmark performance does not necessarily translate into better generation quality. Encoders that preserve spatial structure provide the largest benefit for diffusion training, even when their classification accuracy is lower. This suggests that evaluation protocols for self-supervised encoders should include spatial-coherence criteria when the target application is generative modeling.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) use linear probing to measure the representation gap, using the increase in validation accuracy to explain the effectiveness of REPA. We find that spatial structure metrics (SSM) can also be used to measure the representation gap and explain the improvements across models. As shown in Figure 17, we see that representation alignment helps close the gap between SSM performance of a pretrained encoder like DINOv2 and the diffusion features.

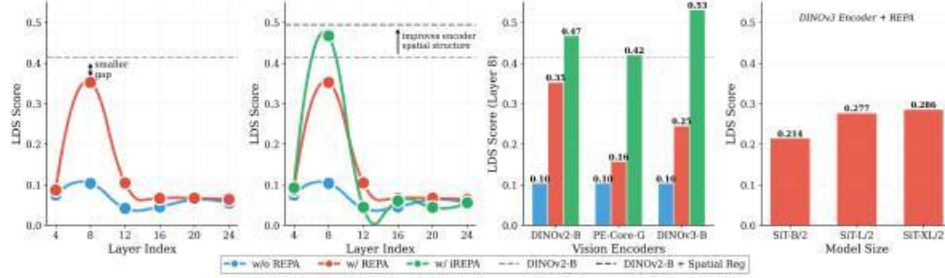


Figure 17: Representation alignment as bridging the spatial feature gap. a) We find that representation alignment can also be seen as bridging the spatial feature gap between diffusion and vision encoder patch features. b) iREPA (§4) accentuates the spatial features of the vision encoder (at cost of some global information) — helping achieve better generation performance. c) iREPA helps consistently improve performance across different vision encoders. d) Spatial structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

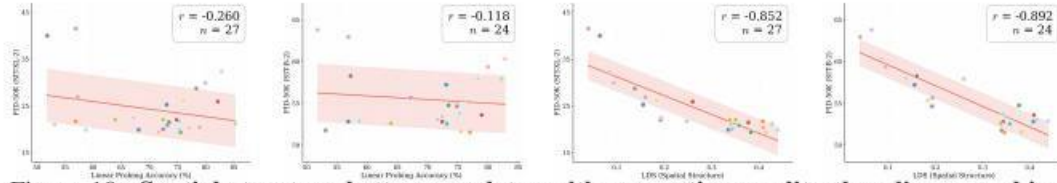


Figure 18: Spatial structure better correlates with generation quality than linear probing. Correlation analysis over 27 vision encoders. **Left two:** Linear probing accuracy versus FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) shows weak correlation. **Right two:** Spatial structure (LDS) versus FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) shows strong correlation. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

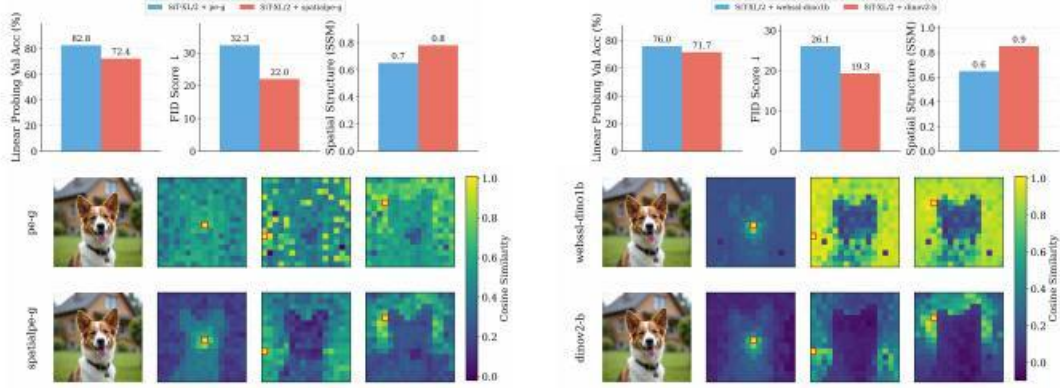


Figure 19: **Motivating examples from recent SSL encoders — spatial structure matters.** **Top row:** Metric comparison showing a mismatch between ImageNet accuracy and generation quality. **Left:** PE-g achieves higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, with much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b shows higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that encoders with better generation (Spatial-PE-g, DINOv2-b) maintain clear spatial coherence, while those optimized for classification (PE-g, WebSSL) lose spatial structure.

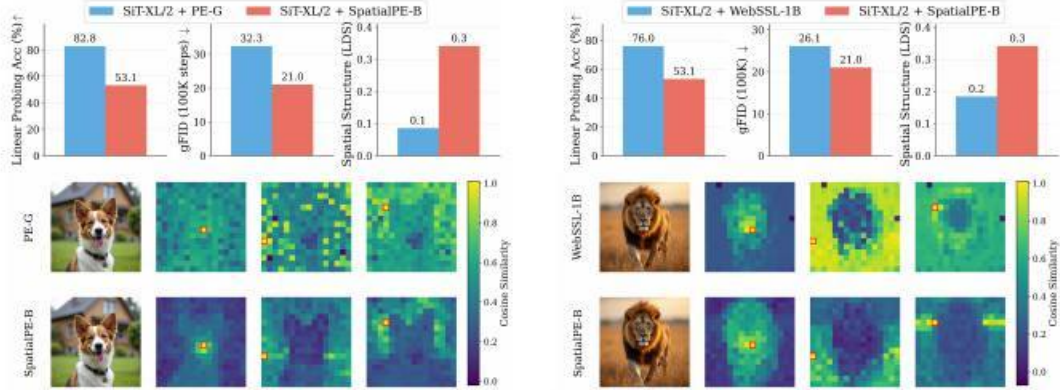


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metric comparison showing a mismatch between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, although it has substantially higher validation accuracy (82.8% vs 53.1%), shows worse generation quality compared to PE_{Spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) also has substantially higher global performance (76.0% vs 53.1%), but worse generation. **Bottom:** Spatial structure provides a more informative predictor of generation quality than global performance. See §3 for the spatial-structure metric. All results are reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize the impact of spatial normalization on vision encoder representations. Vision encoders can have significant global components or overlays that limit the contrast between spatial tokens. For example, tokens in one semantic region can be highly correlated with tokens in unrelated regions (e.g., background), reducing the spatial distinctiveness of features. Spatial normalization removes this global overlay to enhance contrast between different spatial tokens, allowing the model to better preserve local spatial structure while reducing the dominance of global information that can interfere with generation quality.

The following figures demonstrate this effect across different examples, showing how spatial normalization transforms the feature representations to emphasize spatial structure:

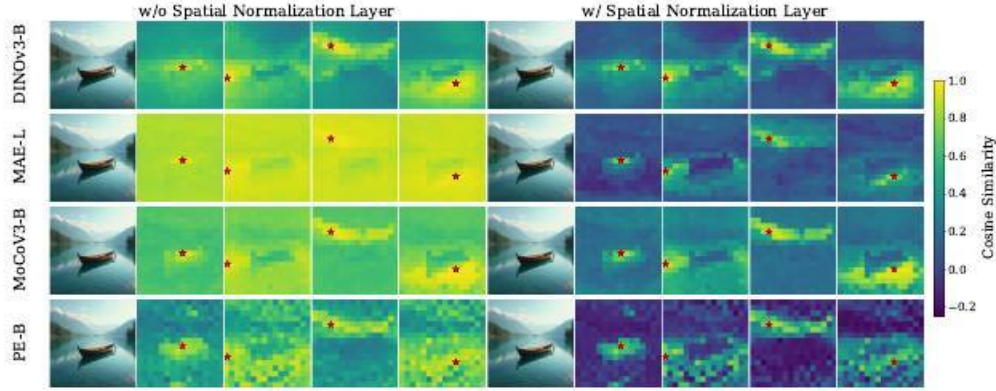


Figure 21: Effect of spatial normalization (Example 1). The heatmaps compare token-similarity patterns before and after spatial normalization. Without normalization (left), global components induce high correlations across unrelated regions. With spatial normalization (right), local spatial structure is strengthened and global interference is reduced, yielding more distinct semantic boundaries.

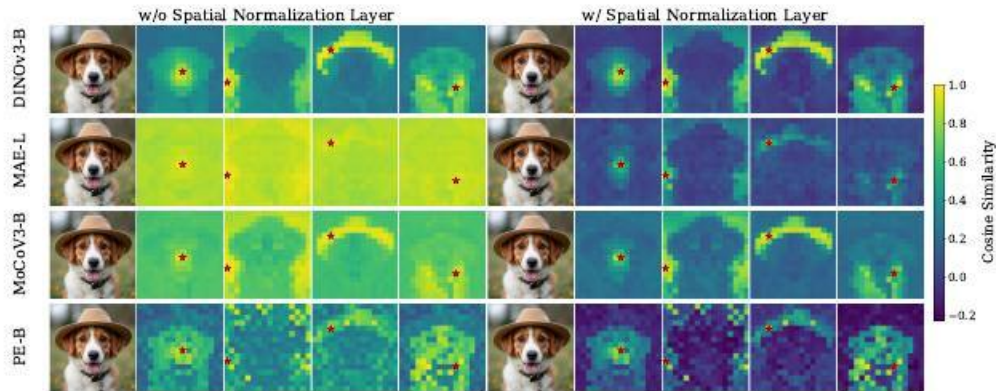


Figure 22: Effect of spatial normalization (Example 3). The heatmaps compare token-similarity patterns before and after spatial normalization. Without normalization (left), global components induce high correlations across unrelated regions. With spatial normalization (right), local spatial structure is strengthened while global interference is reduced, producing more distinct semantic boundaries.

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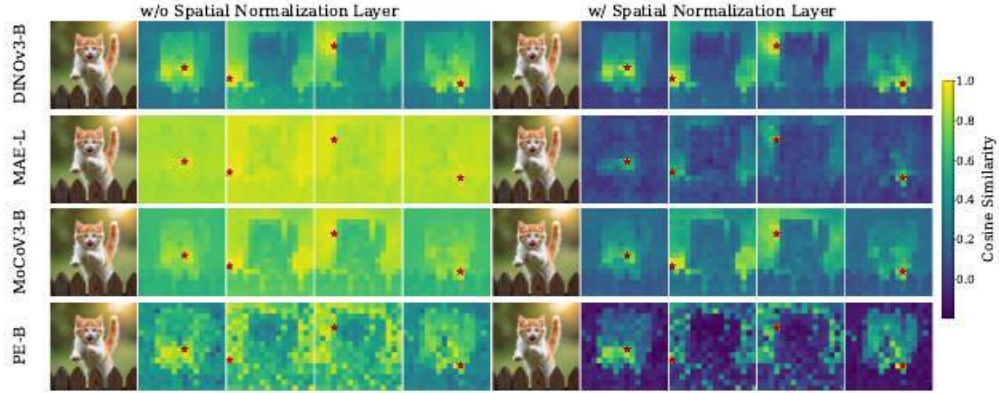


Figure 23: Effect of spatial normalization (Example 2). Feature-similarity maps show that spatial normalization increases spatial contrast. The original features (left) contain a global overlay that reduces separation between foreground and background regions. After normalization (right), spatial tokens become more locally coherent, with clearer separation between semantic regions.

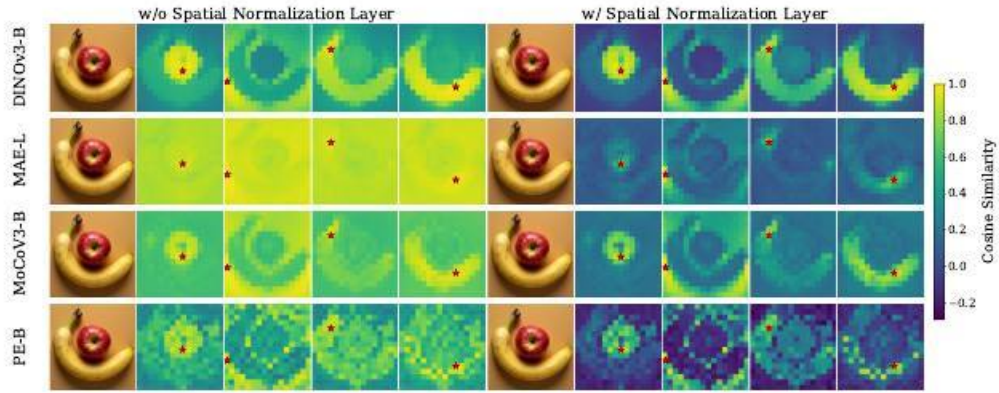


Figure 24: **Effect of spatial normalization** (Example 4). The heatmaps compare token-similarity patterns before and after spatial normalization. Without normalization (left), global components induce high correlations across unrelated regions. With spatial normalization (right), local spatial structure is strengthened while global interference is reduced, producing more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We conduct additional experiments on ImageNet-512 (Deng et al., 2009) to evaluate whether the proposed spatial improvements generalize to higher resolutions. We follow the REPA (Yu et al., 2024) setup and report results across pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.). All results use SiT-XL and SiT-B with 50 NFE at inference and no classifier-free guidance. Fig. 25 reports REPA results before and after applying the spatial improvements (iREPA). The results show that the spatial improvements also generalize to higher resolutions. Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.).

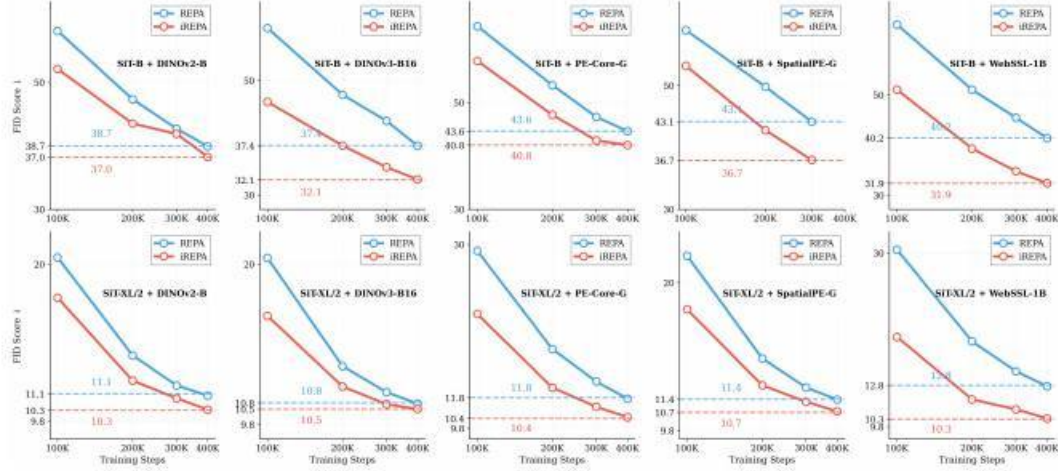


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features helps consistently improve convergence speed across different resolutions for both Imagenet 256 (Figure 7) and Imagenet 512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To further study the generalizability of the proposed spatial improvements beyond ImageNet, we also perform extensive experiments on multimodal T2I tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with various pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). Same as REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256, and evaluated using an ODE sampler with 50 NFE. Fig. 26 shows the results with REPA before and after application of spatial improvements (iREPA). The results show that the spatial improvements also generalize to higher resolutions. Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

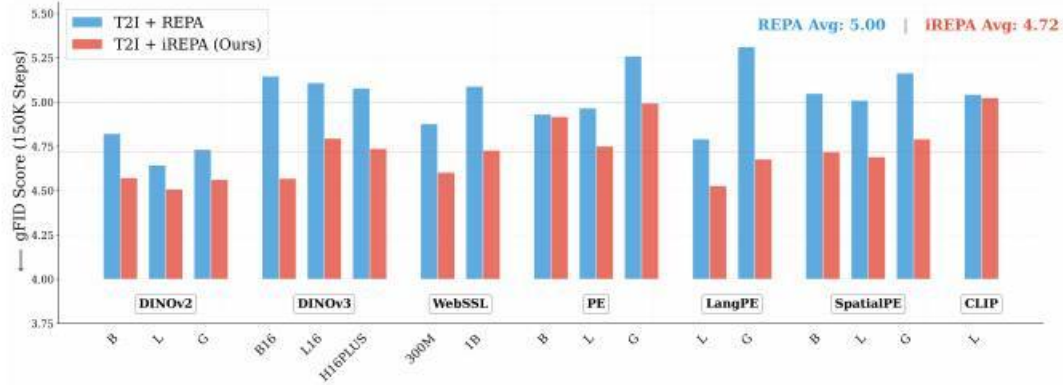


Figure 26: **Text-to-image generation across encoder variants.** Accentuating transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed for both imagenet (refer §4) and multimodal T2I tasks (above). Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 shows the updated correlations across different model sizes. We observe that spatial structure still shows much higher correlation with generation performance over linear probing accuracy. Interestingly, after removing the outliers linear probing actually shows a small positive correlation with gFID (i.e., as linear probing performance increases, the generation becomes worse). This trend is consistent with the observations discussed in Sec. 2, wherein often target representations with higher global semantic performance (linear probing accuracy) show similar or worse generation performance with representation alignment (REPA).

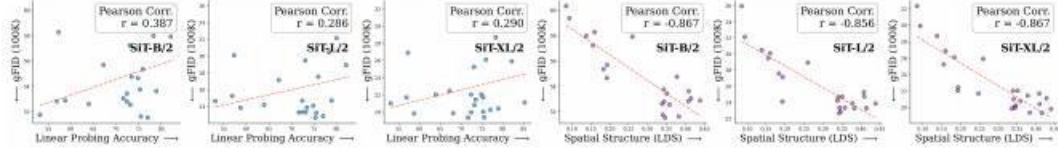


Figure 27: Correlation analysis without outliers. Across all model sizes (B, L, XL) spatial structure still shows much higher correlation (Pearson $|r| > 0.85$) with generation performance over linear probing accuracy (Pearson $|r| < 0.38$) after removing the outliers (MoCOv3-L and MAE-L). Interestingly, after removing the outliers linear probing actually shows a small positive correlation with gFID (i.e., as linear probing performance increases, the generation becomes worse). This trend is consistent with the observations discussed in Sec. 2, wherein often target representations with higher global semantic performance (linear probing accuracy) show similar or worse generation performance with representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after application of spatial improvements with REPA. All results are reported using PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs) and with classifier-free guidance scale of 4.0. Similar to quantitative improvements (§4), we also observe that spatial improvements (iREPA) also help improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

In addition to traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we also verify the robustness of the proposed findings with alternative evaluation metrics for generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). In particular, we use the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation from (Paul, 2024). Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. We next verify the robustness of the key findings from both §3 and §4 with the CMMD metric.

Spatial structure correlates much higher with generation performance than linear probing. To study robustness of analysis from §3, we repeat the large-scale correlation analysis across different vision encoders with CMMD metric (Jayasumana et al., 2024). Results are shown in Figure 29. All results are reported using SiT-B/2 (100K steps) and REPA. All spatial metrics show much higher correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$) — demonstrating that key empirical findings from §3 are robust to the choice of evaluation metric (FID or CMMD).

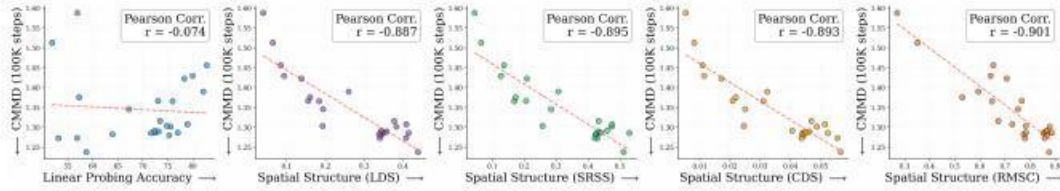


Figure 29: Correlation analysis with CMMD metric. We repeat the correlation analysis from §3 with CMMD metric (Jayasumana et al., 2024) instead of gFID. All results are reported using SiT-B/2 (100K steps) and REPA. All spatial metrics show much higher correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$). This demonstrates that key empirical findings from §3 are robust to the choice of evaluation metric.

Accentuating transfer of spatial features helps improve generation quality. We next study the generalizability of analysis from §4 with the CMMD (Jayasumana et al., 2024) metric in addition to traditional evaluation metrics (gFID, sFID, IS, etc.). Results are shown in Table 8. Across various vision encoders, accentuating transfer of spatial features (iREPA) helps improve convergence speed with both CMMD and traditional evaluation metrics (IS, FID, sFID, Prec., Recall). Furthermore, consistent improvements are observed both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We report additional results using the CMMD metric (Jayasumana et al., 2024). All results use SiT-XL/2, batch size 256, and standard REPA as the baseline. CMMD is computed using the PyTorch implementation from (Paul, 2024), with the standard OpenAI ADM evaluation-suite reference set (Dhariwal & Nichol, 2021) as reference images. Across vision encoders, increasing spatial-feature transfer with iREPA improves convergence speed under both CMMD and traditional evaluation metrics (IS, FID, etc.). This indicates that the main empirical findings from §4 are robust to the evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 study the impact of spatial normalization (§4) on SAM2-S (46M) encoder features². As shown in Table 9, we observe that for SAM2, while spatial normalization layer helps improve performance, the improvements can be marginal. This is because spatial normalization relies on removing the global component (mean of patch tokens) to enhance spatial contrast. Since SAM2 already has little to no global information (validation accuracy < 24%), spatial normalization only slightly improves the spatial contrast (see Figure 30).

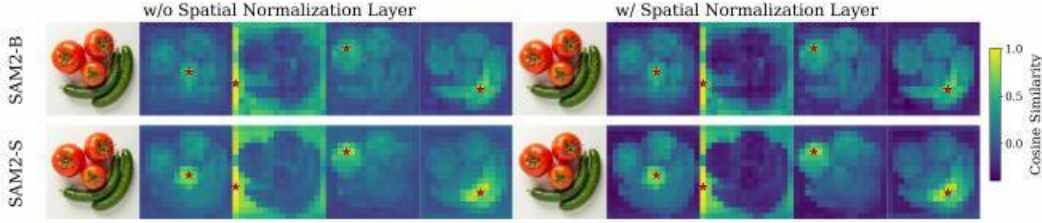


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, the spatial normalization layer increases spatial contrast, but the resulting performance improvements can be marginal. This is because spatial normalization (§4) relies on removing the global component (mean of patch tokens) to enhance spatial contrast. Since SAM2 already has little to no global information (validation accuracy < 24%), spatial normalization only slightly improves the spatial contrast.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** All results use SiT-XL/2 without classifier-free guidance, SAM2-S (46M) as the vision encoder, and standard REPA as the baseline.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 with the validation accuracy after full-finetuning³ instead of linear probing. Results are shown in Fig. 31. All results are reported using SiT-XL/2 (100K steps) and REPA. For full-finetuning validation accuracy, instead of linear probing, we perform full-finetuning of vision encoder with learning rate of $5e-5$, warmup ratio of 0.1, and total of 30 epochs. All spatial metrics show much higher correlation with generation performance than full-finetuning accuracy. Interestingly, gFID actually shows a weak positive correlation with the validation accuracy after full-finetuning (Pearson $r = 0.317$), i.e., as validation accuracy increases, the gfid increases, and generation performance becomes worse. This is consistent with the observations discussed in §2, wherein often target representations with higher global semantic performance show similar or worse generation performance with representation alignment (REPA).

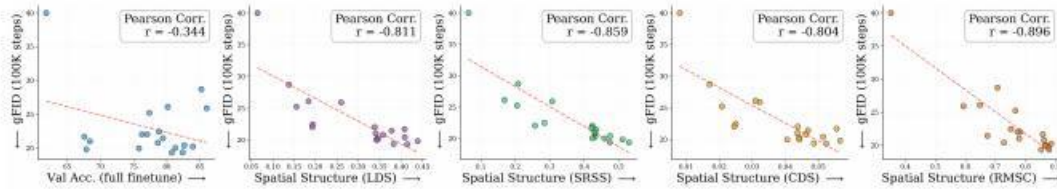


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 using validation accuracy after full finetuning instead of linear probing. All results use SiT-XL/2 (100K steps) and REPA. All spatial metrics show much higher correlation with generation performance than full-finetuning accuracy. Notably, gFID has a weak positive correlation with validation accuracy after full finetuning (Pearson $r = 0.317$), meaning that higher validation accuracy corresponds to higher gFID and worse generation performance. This is consistent with §2, where target representations with higher global semantic performance often show similar or worse generation performance under REPA.

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, iREPA combined with JiT (Li & He, 2025a) outperforms recently proposed state-of-the-art pixel-space diffusion methods such as DeCo (Ma et al., 2025b). All results use 200K training iterations, batch size 256, and 50-step Euler sampling without classifier-free guidance. DeCo results are taken directly from (Ma et al., 2025b).